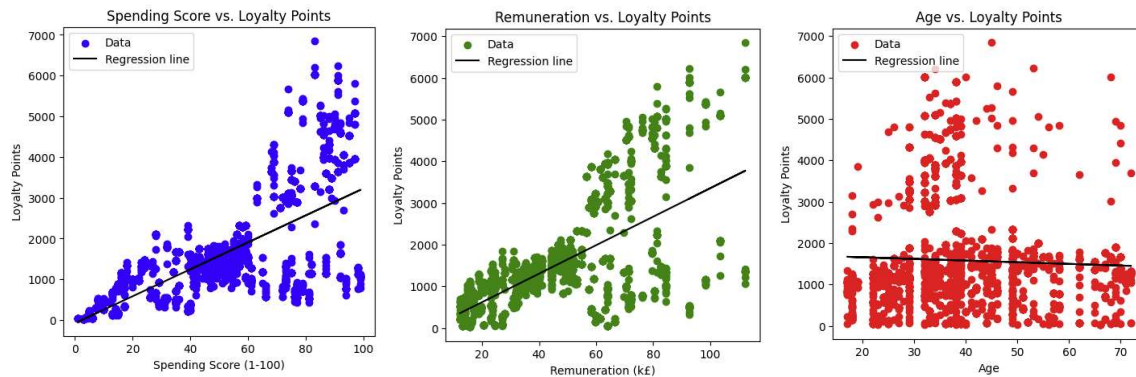


Assignment Report

Corrado Locati

The data were checked, useless columns were dropped and some were renamed. The resulting database was saved.

The linear dependency of the loyalty points with spending score, remuneration and age was calculated and plotted.



By performing the linear regression, quality parameters like adjusted R^2 (the closer to 1, the better the fit) were calculated. The values are 0.452, 0.379, 0.001, for the spending score, remuneration and age, respectively. The loyalty points have a weak linear correlation with the spending score and remuneration, and no linear correlation with the age.

To check which are the important variables for gaining loyalty points, a decision tree was calculated. Categorical variables like education and gender were encoded as numbers, to be incorporated in the decision tree analysis. Then, the data were divided in train and test sets. A decision tree regressor was created with the train data, and the model evaluated.

```
Training R^2 Score: 1.0
Testing R^2 Score: 0.9950294047576341
Mean Absolute Error (MAE): 33.375
Mean Squared Error (MSE): 8051.355
Root Mean Squared Error (RMSE): 89.72934302668219
```

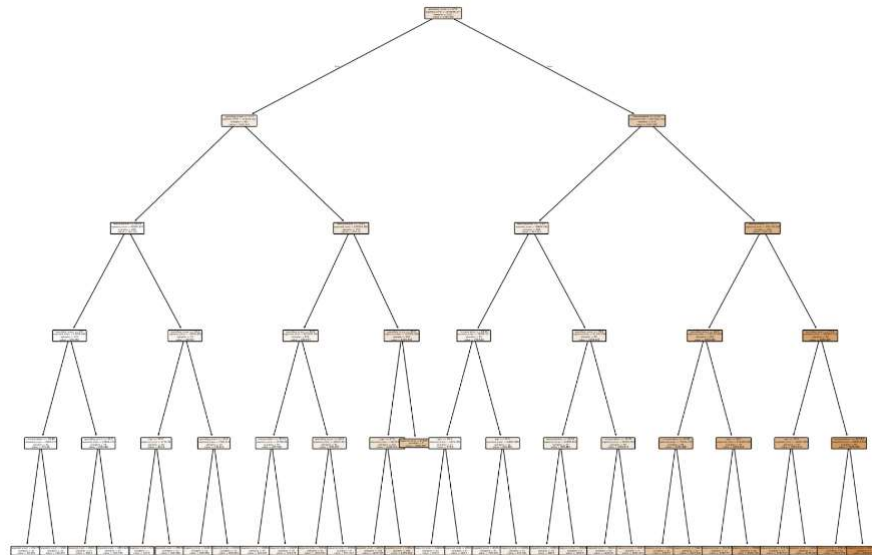
Even if the testing R^2 is high, approx. 99.5%, a decision tree with all the levels (not shown here) was not possible to use, since it was too complicated, and the boxes were overlapping. Therefore, the decision tree

was pruned, with considering only 5 levels of decision, 10 minimum samples to split a node, and 0.01 minimum impurity decrease to make a split. This resulted in a lower but acceptable R^2 of approx. 95.7%.

```

Training R^2 Score: 0.9612192051375839
Testing R^2 Score: 0.9566506653144026
Mean Absolute Error (MAE): 177.11368640533988
Mean Squared Error (MSE): 70217.11999254132
Root Mean Squared Error (RMSE): 264.9851316442893

```



The highest number of loyalty point is reached by people with high spending score and high remuneration. The highest number of loyalty points, on average (6138.143), is reached with a spending score higher than 71.5 and a remuneration higher than 107.830 £. The squared error is high, 83476.408.

```

squared_error = 83476.408
samples = 7
value = 6138.143

```

However, the largest group of customers (498) reaches an average of 1499.466 loyalty points. Such group is older than 27.5 years, with a spending score of 61.5 and a remuneration higher 33620 £. The squared error is high, 108543.61.

```

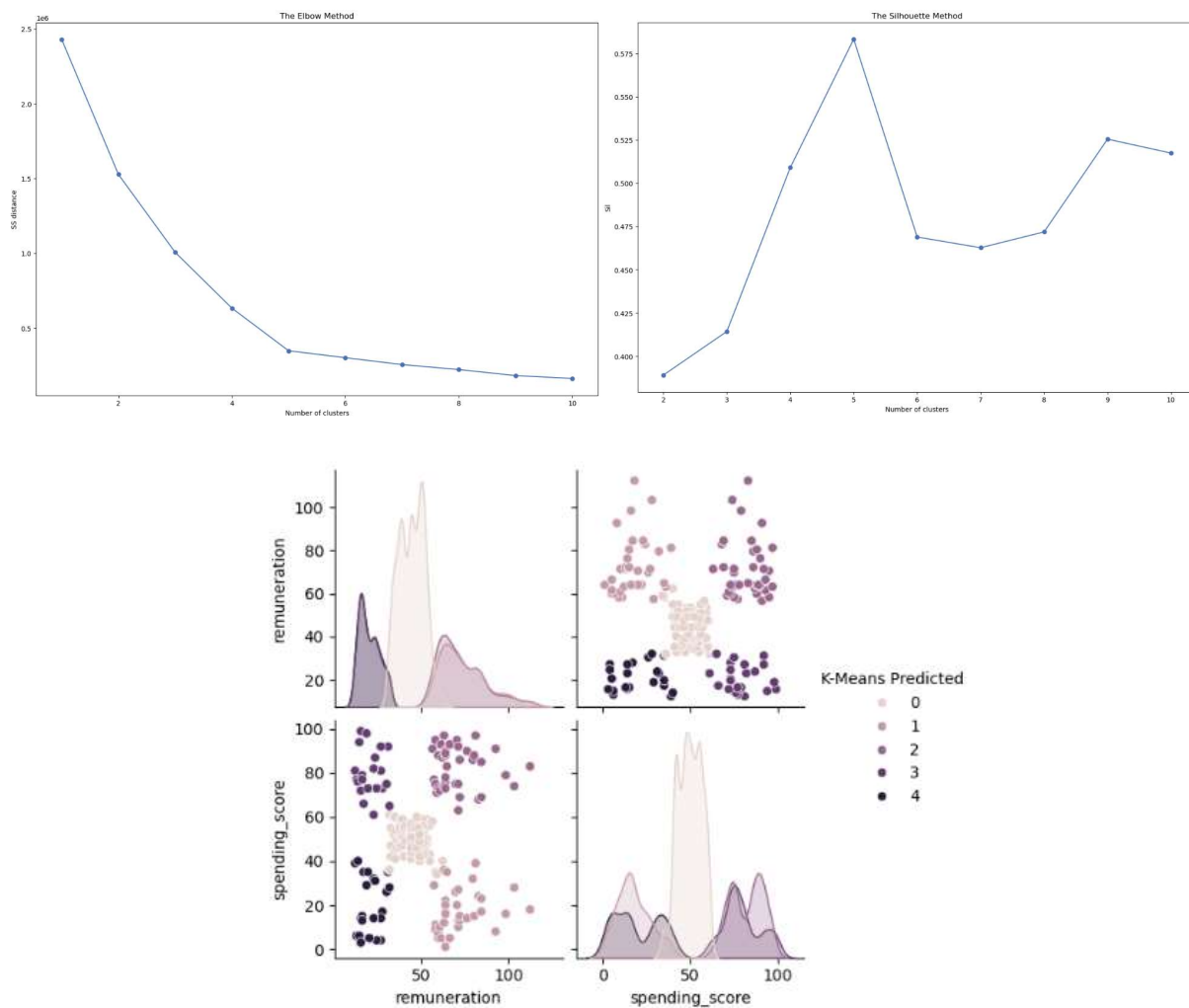
squared_error = 108543.61
samples = 498
value = 1499.466

```

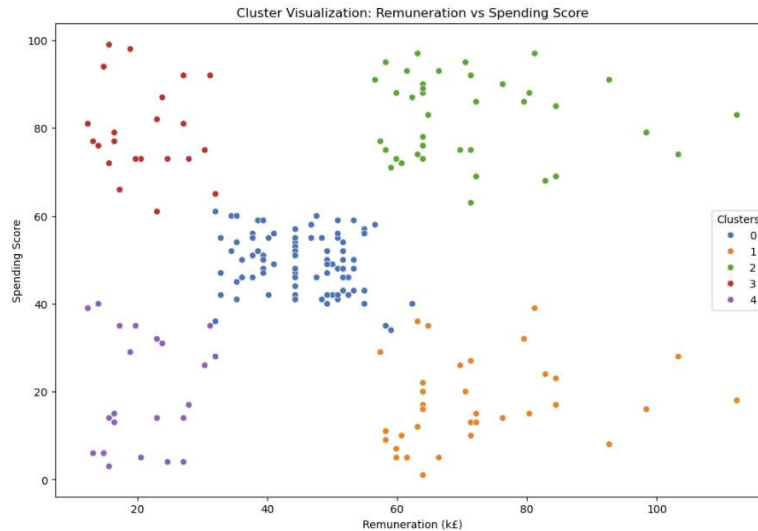
Such high squared errors suggest that a further splitting could be beneficial for a better understanding.

A marketing campaign needs to be targeted towards specific market segments. So, groups within the customer base were clustered looking at their spending score and remuneration. This was accomplished

using k-means clustering. To decide the number of clusters, the Elbow and silhouette methods were used, resulting in a cluster number of five.



The density plots in the diagonal show the distribution of spending score and remuneration within the clusters. The scatter plots (also below) show the distinct clusters, defined by remuneration and spending score values withing a certain range.



To grasp the customer sentiment about Turtle games and their products, products reviews and summary columns were analysed with the natural language processing (NLP) algorithm.

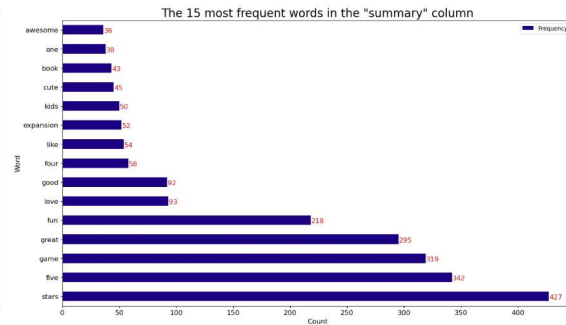
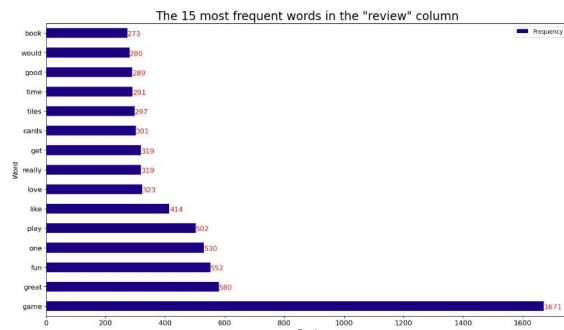
All the words were put in lower case and join with a space, and all the punctuation was removed, as well as duplicates. Rows were tokenised, and word clouds of summary and review were created.



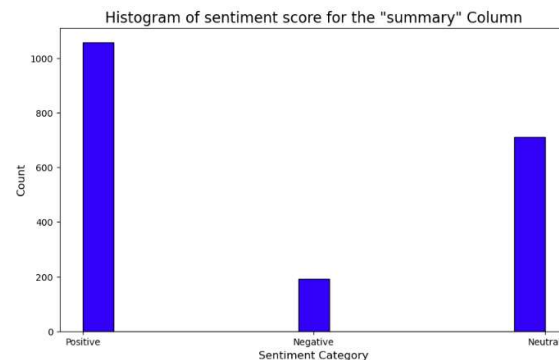
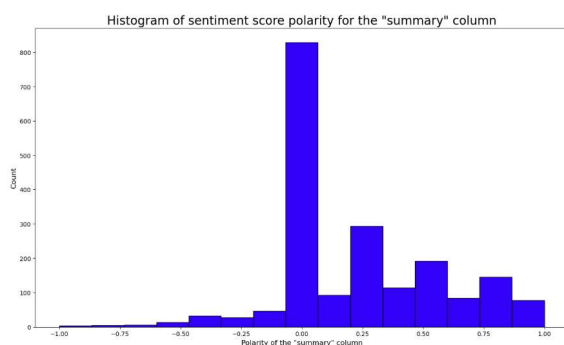
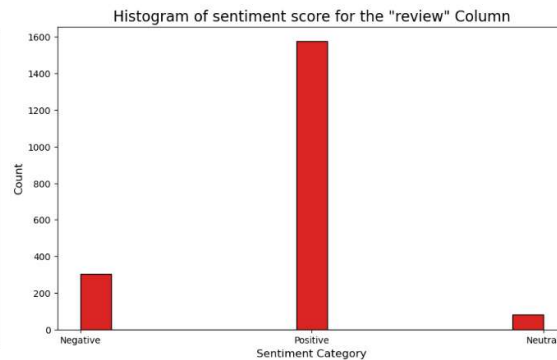
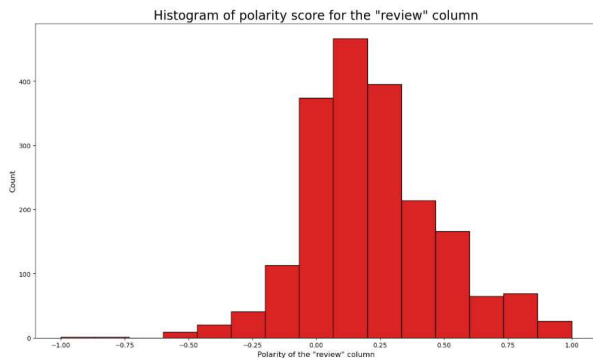
Here, the stopwords were the most common, but they don't give any information. Therefore, stopwords and alphanumeric characters were removed, and new word clouds were created.



The fifteen most common words were collected.



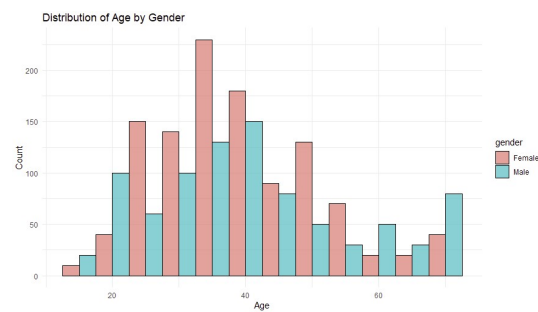
Overall, the most common words are positive in both the columns. Words like game, great, fun, five, stars and so on highlight the goodness of Turtle game work. However, a deeper analysis of the polarity (distribution of words whether they are positive or negative) and of the sentiment (positive, negative, or neutral?) are needed. The results were exposed in histograms.



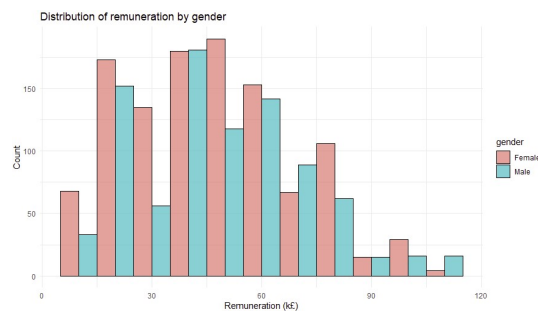
The polarity and sentiment histograms confirm that both reviews and summaries have positive scores. In the reviews, the predominancy of positive words is strong, towards neutral and negative words. In the summaries, there is a high number of neutral words, but still lower than the positive ones.

Further analyses were performed on the data, with R.

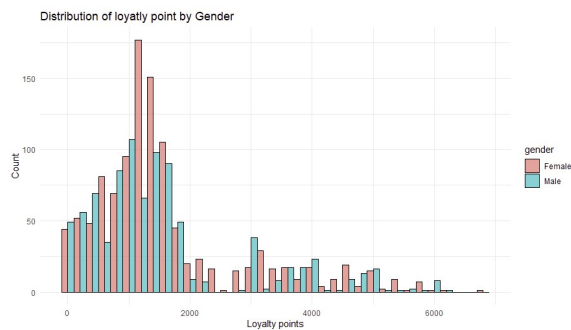
Histograms were plotted to check the distribution of age by gender,



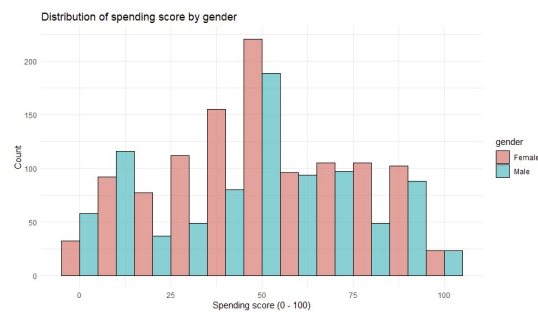
Remuneration by gender



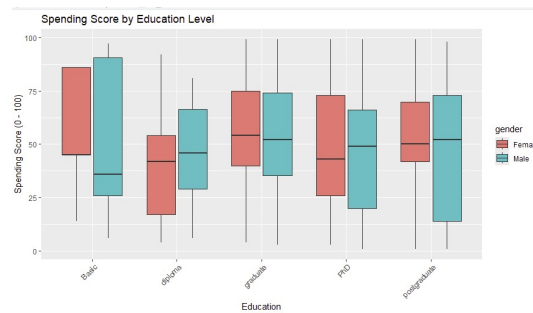
Loyalty points by gender



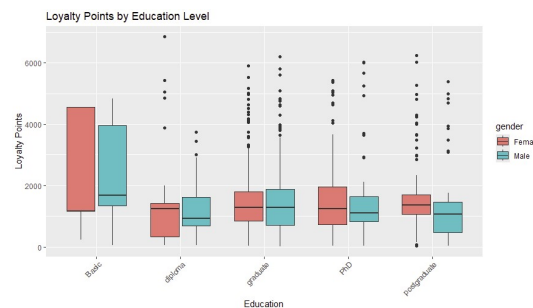
And spending score by gender



Customers are between 19 and 62 years old, with the majority between 30 and 60 years old. Within this range, the females are more than the males. The highest remuneration is between 20K £ and 80K £. The most frequent amounts of loyalty points lay between 0 and 2000. People with a spending score of 50 are higher in number than the others.

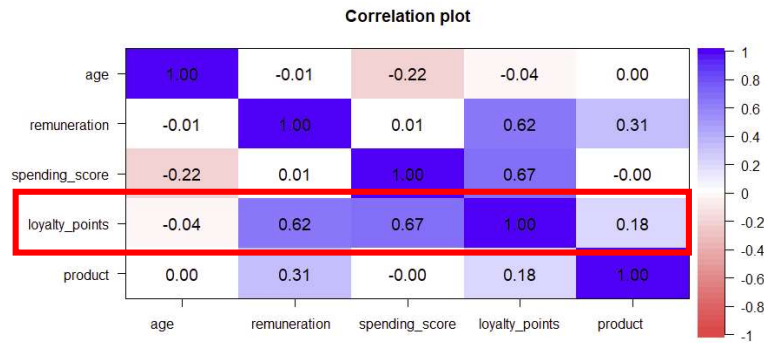


The spending score between genders looks not affected by the education level. The only difference is that male customers with a basic education and with a post graduate diploma spend more than female customers with the same education.



The loyalty points are collected per education level 50% of the customers with basic education collect more loyalty points than the other groups, which have outliers higher than the basic education groups.

A clever way to weight the correlation among the variables is to calculate the correlation matrix. We are interested in the loyalty points, and how strong the other variables correlate with it.



The remuneration and spending score have a quite strong positive correlation with the loyalty points (e.g., the higher the remuneration, the higher the amount of loyalty points collected), and a weak negative correlation with the age. Here, it looks that the age is not an important factor, like deducted from the decision tree.

A multiple linear regression model (MLR) is calculated, with all the descriptive variables. Here the results.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2203.0598	52.3609	-42.08	<2e-16 ***
age	11.0607	0.8688	12.73	<2e-16 ***
remuneration	34.0084	0.4970	68.43	<2e-16 ***
spending_score	34.1832	0.4519	75.64	<2e-16 ***

signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 513.8 on 1996 degrees of freedom
 Multiple R-squared: 0.8399, Adjusted R-squared: 0.8397
 F-statistic: 3491 on 3 and 1996 DF, p-value: < 2.2e-16

It is possible to check which are the important variables by looking at the absolute value of the t-value (the higher, the better), and the Pr (the lower the better). It looks that all the three variables are important, even if before the Age variable was excluded. Such model has an adjusted R^2 of 83.97%.

By removing the age, the model results in:

Coefficients:

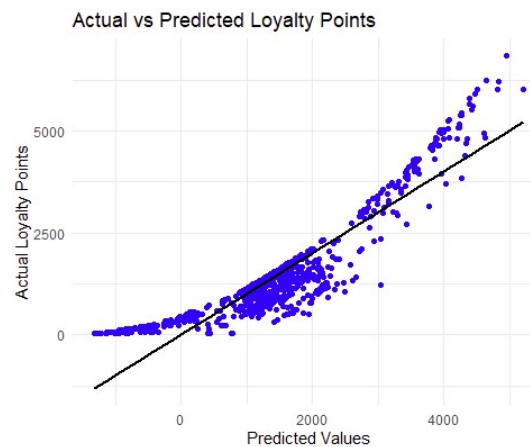
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-1700.3051	35.7396	-47.58	<2e-16	***
remuneration	33.9795	0.5166	65.77	<2e-16	***
spending_score	32.8927	0.4578	71.84	<2e-16	***

 signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 534.1 on 1997 degrees of freedom
 Multiple R-squared: 0.8269, Adjusted R-squared: 0.8267
 F-statistic: 4770 on 2 and 1997 DF, p-value: < 2.2e-16

With a slight decrease of the adjusted R² of 82.67%. This will be the model used.

Plotting the loyalty point vs. the linear regression, should give distribution of points and trendline along a 45° positive slope. Here, the points are not linearly distributed, but spread on a square fashion.



Therefore, the square root of the loyalty points was calculated, and perform a MLR on that.

```
Call:
lm(formula = sqrt_loyalty_points ~ remuneration + spending_score,
    data = turtle_reviews_2_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-18.640	-3.317	1.953	4.007	6.863

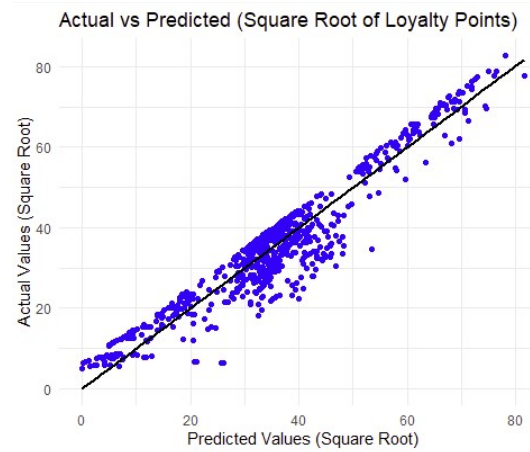
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-4.136673	0.345101	-11.99	<2e-16	***
remuneration	0.403529	0.004989	80.89	<2e-16	***
spending_score	0.426659	0.004421	96.51	<2e-16	***

 signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.158 on 1997 degrees of freedom
 Multiple R-squared: 0.8887, Adjusted R-squared: 0.8886
 F-statistic: 7973 on 2 and 1997 DF, p-value: < 2.2e-16

The t values are high, and the adjusted R², 88.86% is lower than before. The plot shows a linear trend with a 45° slope, meaning that the variables contribute to the square root of the loyalty points.

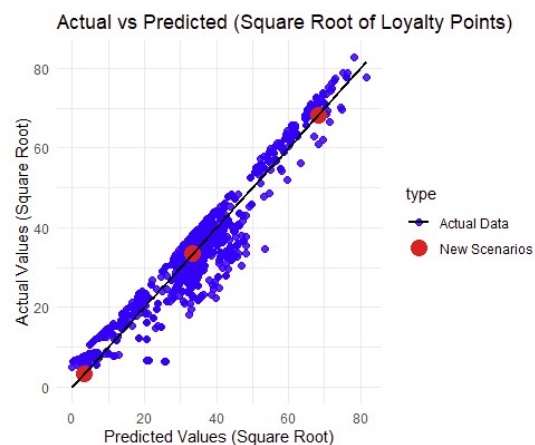


To test the model, three scenarios were calculated.

```
age = c(25, 40, 60),
remuneration = c(15.5, 40.0, 80)
spending_score = c(10, 50, 85)
```

A dataframe with the scenarios was built and merged with the dataframe of actual data.

```
# Combine original data and new scenarios
combined_data <- rbind(
  data.frame(sqrt_predicted = turtle_reviews_2_df$sqrt_predicted,
             sqrt_loyalty_points = turtle_reviews_2_df$sqrt_loyalty_points,
             type = "Actual Data"),
  data.frame(sqrt_predicted = new_scenarios$sqrt_predicted,
             sqrt_loyalty_points = new_scenarios$sqrt_loyalty_points,
             type = "New Scenarios")
)
```



Plotting together with the previous plot shows that the scenarios lay on the trendline, and they are within the data range, proving that the model works.